

Project name: Cornetto: Combinatorial and Relational Network as Toolkit for Dutch Language Technology

Project number: STE05039

Reporting period: 1 October 2007 – 31 March 2008

Participants: Vrije Universiteit, Amsterdam, Faculteit der Letteren (VU-AMS)
Universiteit van Amsterdam, Instituut voor Informatica (UVA-AMS)
K.U. Leuven- ICRI (KU-LEU)
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1 Summary

1.1 Deliverables

The next table lists the deliverables with related work packages, the date of delivery and the status. The numbering is in accordance with appendix-4 of the consortium agreement.

Table 1: Deliverables

Id	Wp	Title	delivery date	status	Mo.
	0	project management en rapportering	continuing	in progress	
D01	1	Cornetto Database	30/09/06	done	6
D02	2	Aligned RBN, DWN and Wordnet2.x	30/09/06	done	6
D03	3	Top-level ontology, relation constraints and assignments	31/12/06	done	9
D04	4	Core Cornetto	30/07/07	done	15
D05	4	Extended Cornetto	31/01/08	in progress	22
D06	5	Acquired concepts	30/03/07	done	12
D07	5	Acquired relations	30/07/07	done	15
D08	5	Cornetto Acquisition toolkit	30/07/07	done	15
D09	6	Ontology for the legal and finance domain	30/03/07	in progress	12
D10	6	Sub-database tuned, trimmed and extended to the legal and finance domain	31/01/08	in progress	22
D11	6	Domain tuning toolkit	31/01/08	in progress	22
D12	7	Evaluation report of the ontology advisory-board	30/03/07	WORKSHOP	12
D13	7	Task-based evaluation of Core Cornetto	30/09/07	in progress	17
D14	7	Task-based evaluation of Extended Cornetto	31/01/08	in progress	22
D15	7	Evaluation report of the user-group on the legal and finance sub-database	31/01/08	WORKSHOP	22
D16	8	Final Cornetto database and software	30/03/08	in progress	24

1.2 Finished deliverables

DEL	WP	Title
D01	1	Cornetto Database
D02	2	Aligned RBN, DWN and Wordnet2.x
D03	3	Top-level ontology, relation constraints and assignments
D04	4	Core Cornetto
D06	5	Acquired concepts
D07	5	Acquired relations
D08	5	Cornetto Acquisition toolkit

1.3 Changes

The editing work will continue until June 2008. The task-based evaluation will take place in June and as well as the completion of documentation and packaging of the results. External evaluation was not budgeted in the project but we will see if we can ask Paola Monachesi to carry out the evaluation for a small fee. Her students already worked with Cornetto.

Instead of an evaluation report by the ontology advisory board (D12) and the user-group (D15), we decided to organize another workshop in September to which we will invite both groups and other interested parties. The previous workshop was very successful and it is a better way to present the results and discuss it then a technical report based on intermediate results.

1.4 Spent effort compared to planned effort

partner	Total person months whole project	Nr. person months spent for period 3	Nr. person months spent in total
VU-AMS	15 (18)*	3 (5.4)	15(21.4)+
UVA-AMS	15	0.8(0)	15(17.9)+
KU-LEU	14	5 (5.2)	10.53 (10.53)
IRION	13 (10)*	3,5 (2.8)	13(16.8)+
Totaal	57		

Comment: the straight number indicate the paid effort in person months, the numbers between brackets give the actual person months including the work exceeding the paid effort.

+ Overspending is based on the overhead of the organization and voluntary contributions.

* Due to the budget shift from VUA to Irion Technologies, the totals have been adjusted.

1.5 Dissemination of the results Cornetto

Papers

Period 4

A. Horák, I. Maks, A. Rambousek, R. Segers, H. van der Vliet, P. Vossen (fc)
["Cornetto Tools and Methodology for Interlinking Lexical Units, Synsets and Ontology"](#), in:
 Proceedings of the 18th International Congress of Linguists ([CIL18](#)), Seoul, Republic of
 Korea, July 21-26, 2008.

A. Pease, C. Fellbaum, P. Vossen, (fc)
["Building the Global WordNet Grid"](#), in: Proceedings of the 18th International Congress of
 Linguists ([CIL18](#)), Seoul, Republic of Korea, July 21-26, 2008.

Vossen P., Fellbaum C. (fc) "Universals and Idiosyncracies in Multilingual WordNets", in:
 Handbook Multilingual Lexicography, [Oxford University Press](#), 2008

Vossen P. (fc) "WordNet: principles, developments and applications", in: Dictionaries. An
 International Encyclopedia of Lexicography. Volume: Recent developments with special
 focus on computational lexicography, Walter/Mouton de Gruyter, [Handbooks of Linguistics
 and Communication Science](#) (HSK), Berlin, 2008

[Vossen P.](#), Maks I., Segers R., VanderVliet H. (fc) ["Integrating lexical units, synsets and
 ontology in the Cornetto Database"](#), in: Proceedings of [LREC 2008](#), Marrakech, Morocco,
 May 28-30 May 2008.

Maks I., P. Vossen, Segers R., VanderVliet H., van Zutphen H. (fc) "Encoding adjectives in
 the Dutch semantic lexical database Cornetto", in: Proceedings of [LREC 2008](#), Marrakech,
 Morocco, May 28-30 May 2008.

Erik Tjong Kim Sang and Katja Hofmann, Automatic Extraction of Dutch, Hypernym-
 Hyponym Pairs. In Proceedings of CLIN-2006, Leuven, Belgium, 2007.

Horak A., P. Vossen, A. Rambousek ["A Distributed Database System for Developing
 Ontological and Lexical Resources in Harmony"](#), in: Proceedings of the 9th International
 Conference on Intelligent Text Processing and Computational Linguistics: [CICLing 2008](#),
 February 17-23, 2008, Haifa, Israel. Also to be published in the [Lecture Notes on
 Computational Linguistics and Intelligent Text Processing in Lectures Notes in Computer
 Science](#), Volume 4919/2008, ISBN 978-3-540-78134-9, 1-15, Springer-Verlag, Berlin, 2008.

Horak A., Vossen P., Rambousek A. (2008) The Development of a Complex-Structured
 Lexicon based on WordNet, in: Proceedings of the Fourth International GlobalWordNet
 Conference - [GWC 2008](#), Szeged, Hungary, January 22-25, 2008

Vossen P., Maks I., Segers R., VanderVliet H., van Zutphen H. (2008) "The Cornetto Database: the architecture and alignment issues", in: Proceedings of the Fourth International GlobalWordNet Conference - [GWC 2008](#), Szeged, Hungary, January 22-25, 2008

Period 3

Vossen P. and C. Fellbaum, 2007 (fc) "Universals and Idiosyncracies in Multilingual WordNets", In: Handbook Multilingual Lexicography, [Oxford University Press](#), 2007

Period 2

Fellbaum C. and P. Vossen, 2007 "Connecting the Universal to the Specific: Towards the Global Grid", In: Proceedings of [The First International Workshop on Intercultural Collaboration](#) (IWIC 2007), Kyoto, Japan, January 25-26, 2007 Publically available on external Website

Vossen, P. (2006). *Cornetto: Een lexicaal-semantische database voor taaltechnologie*, Dixit Special Issue, Stevin. ([PDF](#)) Publically available on external Website

Vossen, P. (2006). *Een communicatief lexicon door het verankeren van woordbetekenissen*, Oratie. ([PDF](#)) Publically available on external Website

Posters

Period 3

Vossen, P., Hofmann, K., de Rijke, M., Tjong Kim Sang, E., and Deschacht, K. (2007). The Cornetto Database: Architecture and User-Scenarios. In DIR 2007, pp. 89-96. ([PDF](#)) Publically available on external Website

Tjong Kim Sang, E. (2007) Extracting Hypernym Pairs from the Web. In Proceedings of ACL 2007 (poster), Prague, Czech Republic.

Period 2

Hofmann, K. and Tjong Kim Sang, E. (2007). Automatic Extension of Non-English WordNets. SigIR 2007, accepted. Publically available on external Website

Talks

Period 4

Guest Lecture on [Corpusgebaseerd tekstonderzoek](#) voor het [ICT](#) Onderwijscentrum van de Vrije Universiteit Amsterdam, December 13, 2007.

Guest Lecture on Corpus-based Methods: "[Automatic term extraction from domain corpora](#)", Universiteit Nijmegen, November 26, 2007

Invited speaker on [Taal, Intelligentie en Betekenis](#)" on the Lustrum symposium [Kunstmatige Intelligentie, Ontwikkelingen en Toepassingen in Muziek, Taal en Geneeskunde](#)" van het [Genootschap Physica](#), Alkmaar, October 27, 2007

Invited Speaker on [Global Wordnet Grid](#) on the 6th International [PLAIN](#) Language Conference, Amsterdam, Netherlands, October 11-14, 2007,

Invited speaker on [Global Wordnet Grid](#) at the [Politechnika Koszalin](#), Koszalin, Poland, October 8, 2007.

Keynote Speaker on [Global WordNet](#) at the [3rd Language & Technology Conference](#): Human Language Technologies as a Challenge for Computer Science and Linguistics, Poznan, Poland, October 5-7, 2007,

Period 3

Tjong Kim Sang, E.: Cornetto presentation on the [Stevin](#) Programmadag, Nederlandse TaalUnie, Hoeven, September 21, Publically available on external Website

Vossen, P, Guest lecture: "Cornetto: Een Combinatorische Semantische Database voor het Nederlands" for the [Capita Selecta](#) of the Instituut voor Nederlandse Lexicologie ([INL](#)), Leiden, May 22, 2007. Publically available on external Website

Vossen, P, Guest lecture: Global WordNet and Global WordNet Grid at the [University of Potchefstroom](#), South Africa, March 19, 2007. Publically available on external Website

Vossen, P, Guest lecture: Cornetto and Wordnets at the [University of Potchefstroom](#), South Africa, March 20, 2007. Publically available on external Website

Vossen, P, Guest lecture: "African Languages WordNet Workshop/Training" of the [Department African Languages](#) of the University of South Africa, (Unisa), Pretoria, South Africa, March 21-23, 2007.

Period 2

Tjong Kim Sang, E.: "Extracting Dutch Hypernymy Pairs from the Web". Presented at CLIN-17, 12 January 2007 in Leuven, Belgium. Publically available on external Website

Vossen, P., Maks, I., Martin, W., van der Vliet, H., Hofmann, K., van Zutphen, H.: *The Cornetto Database*. Presented at CLIN-17, 12 January 2007 in Leuven. ([Abstract](#), [PDF](#), [PowerPoint](#)) Publically available on external Website

Vossen, P.: *Een communicatief lexicon door het verankeren van woordbetekenissen*, Oratie. Presented December 22 2006 at Vrije Universiteit, Amsterdam. ([PDF](#), [PowerPoint](#)) Publically available on external Website

Vossen, P.: *Er kan meer dan men doet met de Cornetto database*. Presented November 30 2006 at the TST Themadag: de gebruiker centraal in Rotterdam. ([PDF](#), [PowerPoint](#)) Publically available on external Website

Period 1

Vossen, P.: *Cornetto*. Presented September 11 2006 at the Stevin project day in Antwerp. ([PDF](#), [PowerPoint](#)) Publically available on external Website

2 Progress per deliverable

2.1 Realized work in the reporting period

WP0: Management

In the 4th period, we had 3 project meetings :

- November, 19th, 2007, Leuven, Consortium and Stevin PC members
- January, 22th, 2008, Szeged, VUA with Masaryk University
- February, 18th, 2008, Amsterdam, Consortium

In addition, we had technical meetings between the individual partners of the project.

For the final period of the project, which lasts 3 months no new project meetings are scheduled.

WP1: Cornetto database

The development of the Cornetto database software was completed during the previous reporting period. In this period, the software was deployed on one additional server, located at VU, for editing work. The server at UvA is used for guest access and is regularly synchronized with VU. Further, minor bug fixes were carried out and user support was provided.

WP2: Aligning semantic resources

Completed

WP3: Ontologize

The ontology assignment was imported from the English wordnet and is further revised and improved for important concepts. This work is parallel to the work done for the editing (see below). Mappings imported from English are either confirmed, corrected or replaced by more

complex mappings. In the case of adjectives, we also started to develop an extension of SUMO.

WP4: Cornetto editing

The editing of the aligned RBN-DWN is carried out for different groups of problematic (e.g. polysemous, frequent) words:

Editing and alignment RBN-DWN-PWN-SUMO-WDOMAIN:

WORDS	period 3	period 4
Adjectives (frequent and / or polysemous):	50	130
Adjectives with 2 senses :	110	110
Frequent verbs:	130	425
Polysemous nouns (4 or more meanings) and nouns with meaning shift labels in RBN:	630	1000
Total	920	1665

The total of 1665 words corresponds with about **9500 lexical units and synsets**. These make up the most frequent, the most complex and most ambiguous words of Dutch. The work carried out consists of mapping RBN to DWN, creating new synsets and lexical units if necessary, adding any other data or relations, mapping the synsets to PWN, assigned a proper mapping to SUMO and a wordnet Domain label.

WP5: Acquisition toolkit

In this period we have adapted one of the online extraction toolkits to the feedback of the lexicographers. We have packaged the required project deliveries for this work package. We have also been working on a conference paper which compares two preprocessing strategies used in our acquisition work.

WP6: Domain acquisition

We have implemented association techniques from the field of data mining among which are a chi-square and a likelihood ratio for a binomial distribution. These tools allow us to detect a variety of events that are correlated such as: terms (including compounds) associated with a domain, collocations associated with a domain, idiomatic expressions, meanings of new words, domain-specific meanings of old words, and the acquisition of horizontal frame patterns.

WP7: Evaluation

Irion implemented 3 methods to implement semantic relations in the classification software:

1. Adding any content word combination (bi-grams) to the index and the classification text;
2. Only create combinations based on the Cornetto database in a window of 15 words;

3. Combined method 1 and 2;

Irion has a Dutch classification system that assigns thesaurus labels from the IPTC thesaurus (<http://www.iptc.org/pages/index.php>) to Dutch news text. We will investigate whether the above methods improve the current system on a test set that is defined for Irion’s customers.

UvA will evaluate the Cornetto database in one of the following end-to-end tasks

- a) Automatic Question Answering (QA) - UvA has developed a QA system with which we are participating in international evaluation efforts. The system currently uses Dutch WordNet for question analysis. The cornetto database can be evaluated by replacing DWN in this step.
- b) Electoral Search - UvA has developed a widely-used electoral search engine verkiezingskijker.nl (<http://verkiezingskijker.nl>). The search engine is currently using manually assigned expansion terms for search categories. Cornetto can be evaluated in the context of automatically selecting such query expansion terms.

Evaluation will be co-ordinated with members of other project groups.

WP8: Distribution

Several users have shown interest in the intermediate results. One user from Princeton University has signed a license for using the intermediate result. Various students have been working with samples from the database and the database is being used by PhD students at the VUA for research in metaphors and opinions in a VICI program. There have also been requests from other project proposals (Stevin and European Union) that intend to use the Cornetto database. The database will also be used in the 7FP KYOTO that started in March 2008 at VUA.

2.2 Problems and solutions

No further problems have been reported

2.3 Suggested planning for the next period

WP3: Ontologize

The final phase of ontologization will focus on

- synsets with many equivalent mappings to English Wordnet
- concepts with high position in the hierarchy
- concepts with conflicting ontology assignments

When the manual work is done for the frequent & polysemous words and for the top concepts, we will extract all inherited SUMO labels for all synsets and check these for conflicts. The conflicts will be traced to their cause and corrected.

WP4: Cornetto editing

Continue editing of problematic cases:

Editing and alignment RBN-DWN-SUMO:

WORDS	Planning period 5	<i>period 4</i>
Adjectives (frequent and / or polysemous)	250	130
Adjectives with 2 senses :	110	110
Frequent verbs:	500	425
Polysemous nouns:	1200	1000
Total	2060	1665

We further focus the editing in the last phase on:

- concepts with many descendants (hyponymy and meronymy);
- concepts with many unmatched LUs and synsets;
- concepts with many equivalence mappings to English wordnet;
- concepts with conflicting SUMO labels and Domain labels;

Finally, we will make sure that all manually edited LUs and synsets will be part of the complete hierarchy in the sense that their hyperonyms are also manually verified. In this way we guarantee a semantic closure of part of the database.

- **Linking DWN-SUMO**

Linking of all aligned synsets (approx. **6000**) to SUMO ontology

- **Compiling guidelines** for linking adjectives, nouns, verbs to the SUMO ontology

MultiWordUnits

- Compiling guidelines for selection, editing, and alignment of MultiWordUnits
- Aligning to DWN, linking to SUMO of a set of 100 MultiWordUnits

WP5: Acquisition toolkit

Work on the acquisition toolkit has been completed. During the remainder of the project period we will publish evaluation / results of this work package.

WP6: Domain acquisition

UvA provides command line version of toolkit with preprocessed data to Leuven and offers support in their initial experiments with the toolkit.

The next step is to perform exhaustive and systematic testing (of which the scenarios have been drafted) of the domain acquisition toolkit. We aim to publish the findings and evaluation in a computational linguistics journal.

WP7: Evaluation

When the final database is delivered in June, Irion and UvA will carry out extrinsic evaluation of the Cornetto database.

WP8: Distribution

We are currently already writing the documentation. We will also carry out evaluation of the quality of the data for different samples. The samples will be drawn from different sets of data based on part of speech, polysemy, mapping-status and revision status. On the basis of these samples, we can assign a quality label to the data records in the total database.